

PHATTHAYA, Thailand

WMO: 484610

Lat: 12.9233N

Lon: 100.8657E

Elev: 61

StdP: 100.60

Time Zone: 7.00 (E07)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 21.0	(c) 22.1	(d) 12.5	(e) 9.1	(f) 23.2	(g) 14.3	(h) 10.2	(i) 24.0	(j) 10.0	(k) 25.6	(l) 8.7	(m) 25.6	(n) 4.5	(o) 40	(p) 0.372

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 5	(b) 4.7	(c) 33.5	(d) 26.8	(e) 32.9	(f) 26.7	(g) 32.4	(h) 26.5	(i) 27.9	(j) 31.9	(k) 27.6	(l) 31.5	(m) 27.2	(n) 31.0	(o) 3.4	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 26.8	(b) 22.6	(c) 30.3	(d) 26.5	(e) 22.2	(f) 30.0	(g) 26.2	(h) 21.8	(i) 29.7	(j) 89.6	(k) 32.0	(l) 88.0	(m) 31.6	(n) 86.8	(o) 31.1	(p) 29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.2	(b) 6.7	(c) 5.9	(e) 19.2	(f) 34.8	(g) 1.2	(h) 1.5	(i) 18.4	(j) 35.9	(k) 17.7	(l) 36.8	(m) 17.0	(n) 37.6	(o) 16.2	(p) 38.7		
(a) 8.2	(b) 6.7	(c) 5.9	(e) 14.6	(f) 28.5	(g) 1.5	(h) 0.5	(i) 13.5	(j) 28.8	(k) 12.6	(l) 29.1	(m) 11.8	(n) 29.4	(o) 10.7	(p) 29.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	28.2	26.6	27.7	28.6	29.7	29.7	29.2	28.8	28.7	28.1	27.7	27.5	26.6	(6)
(7)		DBStd	1.61	1.41	1.11	1.25	1.27	1.24	1.08	1.07	0.93	1.06	1.00	1.34	1.84	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	6658	514	495	578	592	610	577	583	580	543	548	524	514	(10)
(11)		CDD18.3	3617	256	262	319	342	352	327	324	322	293	289	274	256	(11)
(12)		CDH23.3	41185	2326	2726	3738	4358	4561	4129	3941	3863	3300	2989	2832	2421	(12)
(13)		CDH26.7	15228	583	835	1413	2020	2132	1787	1537	1457	1084	877	846	657	(13)
(14)	Wind	WSAvg	2.1	2.0	1.9	2.0	1.8	1.9	2.2	2.5	2.4	1.7	1.6	2.5	2.6	(14)
(15)	Precipitation	PrecAvg	1187	21	19	54	70	138	145	109	107	226	230	55	11	(15)
(16)		PrecMax	1565	78	68	188	197	273	334	242	254	404	387	175	43	(16)
(17)		PrecMin	888	0	0	2	5	32	21	18	42	120	90	0	0	(17)
(18)		PrecStd	175	24	18	44	46	68	73	62	54	74	76	44	13	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.9	32.5	33.2	34.3	34.2	33.1	32.5	32.5	32.1	32.1	32.6	32.1	(19)
(20)			MCWB	23.4	24.3	26.0	27.2	27.5	26.8	26.2	26.2	26.3	26.2	25.7	25.0	(20)
(21)		2%	DB	30.8	31.5	32.4	33.6	33.4	32.4	31.8	31.7	31.4	31.3	31.6	31.1	(21)
(22)			MCWB	24.1	24.8	26.2	27.0	27.2	26.6	26.1	26.1	26.2	26.1	25.6	25.0	(22)
(23)		5%	DB	30.1	30.9	31.8	33.0	32.7	31.9	31.3	31.2	30.9	30.8	30.9	30.5	(23)
(24)			MCWB	24.2	25.0	26.2	26.9	27.0	26.4	26.0	26.0	26.0	26.0	25.6	24.8	(24)
(25)		10%	DB	29.5	30.3	31.3	32.4	32.1	31.3	30.8	30.7	30.3	30.2	30.3	29.8	(25)
(26)			MCWB	24.1	25.0	26.0	26.8	26.8	26.3	25.9	25.9	25.9	25.9	25.2	24.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.4	26.9	27.7	28.3	28.5	27.7	27.2	27.3	27.5	27.5	27.5	27.1	(27)
(28)			MADB	29.6	30.6	31.7	32.9	32.7	31.4	30.6	30.7	30.6	30.7	30.6	30.4	(28)
(29)		2%	WB	25.9	26.2	27.1	27.8	28.0	27.3	26.8	26.9	27.0	27.0	26.9	26.3	(29)
(30)			MADB	29.0	29.9	31.1	32.2	31.9	30.9	30.3	30.3	30.2	30.3	30.2	29.6	(30)
(31)		5%	WB	25.5	26.0	26.7	27.5	27.6	27.0	26.6	26.6	26.6	26.6	26.4	25.8	(31)
(32)			MADB	28.5	29.6	30.5	31.7	31.4	30.6	30.0	29.9	29.7	29.7	29.5	29.0	(32)
(33)		10%	WB	25.1	25.7	26.4	27.1	27.2	26.7	26.2	26.3	26.2	26.2	26.0	25.3	(33)
(34)			MADB	27.9	29.0	30.0	31.1	30.9	30.3	29.7	29.6	29.2	29.1	28.9	28.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.8	5.5	5.1	5.2	4.7	4.3	4.0	4.1	4.4	5.1	5.3	5.7	(35)
(36)			MCDBR	6.4	6.0	5.6	5.5	5.1	4.7	4.4	4.5	4.8	5.7	6.0	6.1	(36)
(37)			MCWBR	3.0	2.4	1.9	1.9	1.9	1.9	1.8	1.8	2.0	2.5	2.7	3.1	(37)
(38)		5% WB	MCDBR	5.1	5.1	5.2	5.2	5.0	4.6	4.2	4.3	4.7	5.4	5.4	5.4	(38)
(39)			MCWBR	2.3	1.9	2.0	2.1	2.2	2.0	2.0	2.0	2.3	2.7	2.6	2.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.532	0.585	0.596	0.540	0.476	0.455	0.452	0.453	0.465	0.540	0.501	0.486	(40)	
(41)		taud	2.014	1.896	1.910	2.092	2.325	2.393	2.397	2.380	2.331	2.080	2.173	2.174	(41)	
(42)		Ebn at Noon	771	747	750	789	827	835	840	850	845	775	795	802	(42)	
(43)		Edh at Noon	172	200	201	167	130	120	120	124	130	164	145	144	(43)	
(44)	All-Sky Solar Radiation	RadAvg	5.03	5.48	5.59	5.90	5.33	5.08	4.91	5.01	4.85	4.75	4.91	4.91	(44)	
(45)		RadStd	0.32	0.27	0.43	0.38	0.36	0.26	0.28	0.33	0.28	0.31	0.37	0.26	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (29 neighbors)	N/A	N/A	N/A	N/A	+0.20	+0.20	N/A	N/A	+72	+72

Nomenclature: See separate page